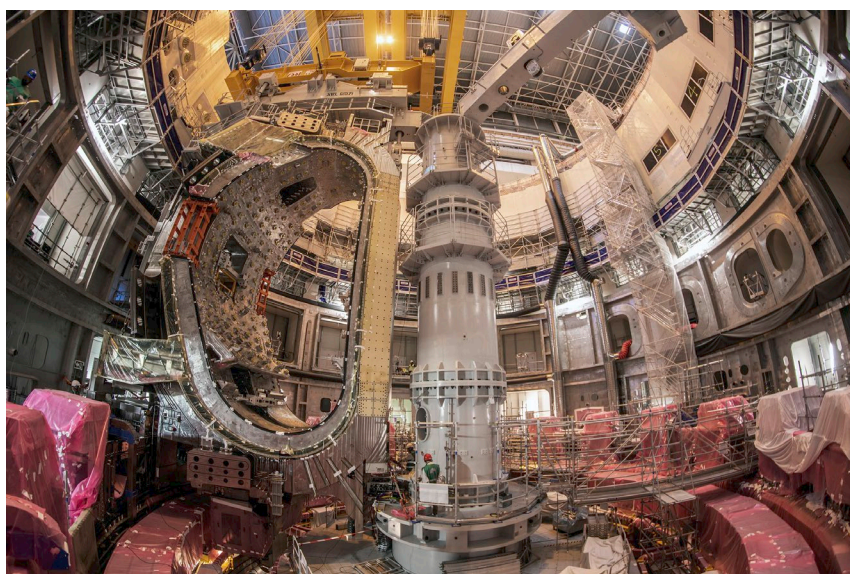

Fusion energy: A paradigm shift in power generation for Europe?



IN-DEPTH ANALYSIS

Panel for the Future of Science and Technology



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Authors: Vasco Guedes Ferreira and António Vale
Scientific Foresight Unit (STOA)

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Fusion energy: A paradigm shift in power generation for Europe?

Fusion, regarded by many as the ultimate form of clean energy, may finally be within reach. This analysis looks at the context of the latest advances, in particular the explosion of private-sector interest. Overall, this has led to an acceleration of many planned timelines, although a lot of uncertainty remains, given the magnitude of the remaining technical challenges. In addition, questions also persist around skills, the regulatory environment, financing and economic viability. With a new EU strategy expected in late 2025, this analysis concludes with an overview of current policy, and discusses some of the challenges the EU faces in setting a pathway towards the commercialisation of fusion energy.

AUTHOR(S)

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ADMINISTRATOR RESPONSIBLE

Vasco Guedes Ferreira and António Vale, Scientific Foresight Unit (STOA)

To contact the publisher, please e-mail stoa@ep.europa.eu

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Executive summary

Fusion has long been the holy grail of energy research. It is deemed a safe, clean, dispatchable (available on demand) and virtually limitless source of power, with widely abundant fuel, which does not share the main risks of nuclear fission. As such, it could play an important role as a form of clean power in a future energy mix in the European Union (EU), supporting the decarbonisation and electrification of the economy, while strengthening European energy autonomy.

The EU has historically been a strong backer of fusion research, reflected in its support for the Joint European Torus (JET) and International Thermonuclear Experimental Reactor (ITER), the multinational experimental reactor under construction in France. In the conventional EUROfusion roadmap, the DEMO pilot plant should follow in the 2050s. However, over the last few years this scenario has been disrupted by the emergence of private start-ups, mostly in the United States, promising to deliver electricity as early as the mid-2030s, backed by substantial investment and pursuing a multitude of approaches. Some public plans are also adapting to this more ambitious timeline, with, for example, Germany, the United Kingdom, the US and Japan all aiming for an operational pilot plant by 2040.

Despite this optimism, significant uncertainty remains, particularly around plasma control, the fuel cycle and materials, and fusion in general is commonly seen as being at low technology readiness level. In addition, there are challenges related to regulation, availability and training of skilled labour, as well as financing. The sustainability of the ITER supply chain will also depend on future fusion initiatives. Furthermore, the economic case for fusion is still unclear as costs are not yet fully known.

The European Commission has committed to presenting a fusion strategy in 2025. Besides clarifying the pathway towards commercialisation, it is widely expected to present a proposal for future public-private partnerships to coordinate efforts between the public and private sectors, including support for fusion start-ups and revised forms of governance, as well as the future approach to regulation. It will likely also present a revised fusion roadmap, possibly with more ambitious goals, similarly to Japan's recent plans. Finally, the strategy may also include proposals on the approach to international collaboration and intellectual property (IP) ownership, as well as plans to sustain the fusion supply chain and ensure availability of skills.

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1. Introduction

Fusion powers the sun and the stars. If replicated in a controlled fashion on Earth, fusion is a potential source of clean energy, which can significantly decarbonise the economy and improve energy security. Unlike nuclear fission (the technology used in conventional nuclear power plants and small modular reactors), which involves splitting heavy atoms to release energy, fusion combines light atoms to form a heavier one, releasing vast amounts of energy in the process. This fundamental difference makes fusion a more attractive option, as it does not produce long-lived radioactive waste or present the risk of nuclear disasters like Fukushima or Chernobyl.

The case for fusion is compelling, and multilateral international research collaboration has been ongoing for several decades, as exemplified by the Joint European Torus (JET), and more recently the International Thermonuclear Experimental Reactor (ITER). However, achieving a sustained and energy-positive fusion reaction has proved difficult, with unsolved technical challenges and high investment costs. Nevertheless, the potential benefits make it an attractive option for future energy systems, in particular from an energy security and strategic autonomy point of view. Firstly, fusion has the potential to be a nearly limitless source of energy because the fuel used in fusion reactions is widely available. Secondly, fusion is a low-carbon energy source, producing no greenhouse gas emissions. Thirdly, fusion energy has the potential to be a reliable source of energy, making it a potentially interesting complement to intermittent renewable energy sources like solar and wind power.

Achieving commercial fusion energy will require significant technical and engineering advances and investment. The scale of the challenge is encapsulated in the ambition to build a machine that contains a plasma ten times hotter than the sun, while magnets colder than outer space sit just outside. Public and private organisations in the EU, US, China and elsewhere are working on addressing these issues and testing different fusion energy technologies.

Fusion has long been the subject of a joke that it was '30 years away and would always be'. However, the field has undergone significant changes in the past few years. Technological advances, for example in artificial intelligence (AI) and high-performance computing, or the development of high-temperature superconductors, are enabling important progress in reactor design and plasma control. In turn, this raises the possibility of faster and cheaper pathways towards commercial reactors, having as consequence a recent explosion of interest in fusion from the private sector, with multiple start-ups appearing worldwide, exploring different pathways for fusion and raising significant sums.¹ They also promise very ambitious timelines, with some claiming they will be injecting electricity into the grid already within the next decade, at considerable odds with the main ITER roadmap, which sees this happening only in the second half of the 21st century.

¹ See discussion in National Academies of Sciences, Engineering, and Medicine, [Final report of the Committee on a strategic plan for US burning plasma research](#), The National Academies Press, 2019, particularly chapter 4 ('Advancing magnetic fusion toward an economical energy source'). It is easy to see the findings and recommendations as inspiration not just for the current US strategy, but also the approach of some start-ups, in particular commonwealth fusion systems (note mentions of an ARC (affordable, robust, compact) device).

2. Fusion basics

To think about fusion, you can start with a simple question: what holds an atomic nucleus together? Two forces are at play: protons, which are charged, repel each other through the electromagnetic force, but the strong nuclear force binds neutrons and protons together. When light nuclei fuse, the strong nuclear force means the resultant nucleus is more tightly bound, with a smaller mass, and energy is therefore released in the process – according to Einstein's famous equation. However, this strong nuclear force has a very short range, and therefore to achieve fusion it is necessary to first overcome the electromagnetic repulsion and bring the nuclei close enough, strongly enough, and for a sufficiently long time. This is encapsulated by the Lawson criterion, which looks at fusion conditions through a triple product of density, temperature and confinement time, and which can be related to the threshold necessary to achieve sustainable fusion reactions.² Lacking the sun's immense gravity, fusion reactors rely on very high temperatures to achieve fusion. At these temperatures, matter is in a state of plasma and, consequently, electrically charged. This has very important consequences, as it means it can be contained using magnetic fields but can also be subject to coordinated motions over large scales. Plasma instabilities and the difficulty of controlling plasmas at these very high temperatures have been among the biggest barriers to the development of fusion technologies.

2.1. Fuels

The choice of fuel mix for achieving fusion has significant implications. This choice has several important impacts at the level of reactor design, including the required temperature, neutron production and fuel availability. The following fuels have all been proposed by at least one of the public or private fusion developers:

- Deuterium-tritium: this is the fuel mix most widely chosen, as it achieves the highest reaction rates at lowest temperature and therefore presents lowest technical barriers. Deuterium and tritium are both heavier isotopes of hydrogen, but while deuterium is abundant in seawater, tritium does not occur in nature as it decays in around 12 years and must be bred in the reactor via fission of lithium, taking advantage of the neutrons produced in the fusion reaction. These energetic neutrons also require special care to be paid to shielding and remote maintenance of the reactor. This also leaves some reactor parts radioactive, posing a challenge for decommissioning, although unlike with nuclear fission reactors the radioactivity is not long-lived.
- Deuterium-deuterium: as a fuel choice, deuterium presents two key advantages over a deuterium-tritium mix; it avoids the problem of tritium breeding, and the reaction produces less energetic neutrons, which are therefore easier to handle. However, it requires a higher temperature to reach similar reaction rates. Research reactors often run on this fuel, since many do not have energy break-even as a primary goal.
- Deuterium-Helium-3: This mix produces fewer neutrons and allows for direct electricity production through charged particles but requires yet again a higher temperature. Helium-3 is rare on Earth, but exists in abundance on the moon, so this mix may be better suited to space-based reactors.
- Proton-Boron-11: the key advantage of this mix is the fusion reaction produces charged particles but no neutrons, and therefore direct electricity production is possible while requiring minimal shielding and producing little to no waste. It faces

² See S. E. Wurzel and S. C. Hsu, '[Progress toward fusion energy breakeven and gain as measured against the Lawson criterion](#)', *Phys. Plasmas*, Vol. 29 (062103), AIP Publishing, 2022, for a technical explanation and historical comparison.

significant technical barriers, as the temperatures required are an order of magnitude greater than for deuterium/tritium.

2.2. Reactor design

Equally, different approaches can be taken to designing reactors when it comes to creating the necessary conditions for fusion and controlling the resulting plasma. They can essentially be divided into two broad categories: magnetic confinement (MCF) and inertial confinement (ICF):

- In magnetic confinement, the plasma is usually kept inside a toroidal-shaped chamber (although it can also be spherical) using magnetic fields. In general, MCF devices suffer from issues of plasma instability, which can cause significant heat discharge onto the reactor walls, leading to energy loss in the plasma, thereby affecting fusion efficiency. This, combined with neutron bombardment in the fuel mixes most often used, leads to degradation and loss of structural integrity in the inner walls, posing considerable technical and maintenance issues. Furthermore, for efficient operation very powerful magnetic fields are required, necessitating expensive superconductor magnets that must be cooled cryogenically. Despite these important technical barriers, MCF has been the most widely pursued approach, given it presents the clearest route to steady-state operation, with for example tokamak-type devices being used in public research for several decades. More recently, an alternative to the simpler doughnut-shaped tokamak has emerged – the stellarator, which uses a more complex twisted structure. While the latter is more challenging to design, requiring advanced computer simulations, in operation it offers better and easier plasma stability.
- Unlike magnetic confinement, which seeks to operate the plasma in a steady state, inertial confinement works in short pulses, with inertial forces temporarily ensuring the necessary confinement for fusion to occur. The most typical route is to use lasers to implode a small fuel pellet – this is the approach used at the National Ignition Facility (NIF) in the US, which holds the current record in energy gain from fusion. The two other pathways being explored involve a compression either through magnetic forces via strong electrical currents, or through some kind of mechanical force. In terms of power production, the key challenge of ICF is its pulsed nature, in the sense that a high pulse frequency would be required, which currently remains theoretical. There are also technical barriers in the complexity of the symmetry required, as well as the progressive degradation of the compression mechanisms, given the high stresses involved.

There is in addition a third category, which forms something of a hybrid, usually called magneto-inertial (MIF) approach, which follows a similar concept to magnetic confinement, combined with magnetic compression of the plasma. Field-reversed configurations (FRC), where part of the necessary magnetic fields is formed by currents within the plasma itself, often feature in MIF designs, although they can also use magnetic confinement alone. The main advantage of this category is that it achieves higher densities than in MCF, opening the door to the use of different fuel mixes, while having a simpler geometry as well as avoiding some of the complexity of ICF. This is the least advanced area of research, mostly being pursued by private actors, and faces significant barriers in terms of plasma stability and control.

2.3. Net energy gain

An important concept in fusion energy debate, and which requires some care, is the idea of net energy gain, usually expressed by the letter Q . However, this simple concept can be deceiving, as

when it comes to fusion, energy in and energy out can be difficult to define: considerable energy is needed to heat the plasma and achieve fusion conditions, much of it might be wasted, and also when used for other purposes such as cryogenics for superconducting magnets. Equally, there will be efficiency losses on the output side as most fusion reactions produce heat, which must then be turned into electricity. Most discussions of Q ignore these factors, and look only at the 'scientific ratio' between the energy released in the fusion reactions and that used to heat the plasma. So, for example when the NIF announced it had achieved the first net (scientific) energy gain in late 2022, with a Q greater than one, this accounted only for the energy actually reaching the target fuel, with the electricity needed to power the experiment being a couple of orders of magnitude greater. While achieving a Q of 1 remains a significant scientific milestone, actual commercially viable energy gain likely requires a Q approaching 20 to 30.³

3. The path to fusion

Developing fusion energy technology is a complex and challenging process requiring significant investment. Several public and private initiatives are underway to demonstrate the feasibility of fusion energy and pave the way for its commercialisation. However, stakeholder projections appear to put the timeline for commercial fusion anywhere between 10 years and several decades away.⁴

3.1. Current outlook

Originally planned as, and to this day, a key project in the development of fusion energy, ITER is an international collaborative project (EU, China, Japan, India, the Republic of Korea, the Russian Federation and the USA). The project aims at demonstrating the scientific and technological feasibility of fusion energy to achieve a plasma state that can sustain a fusion reaction for a significant amount of time. ITER is currently under construction in Cadarache, Saint Paul-lez-Durance, southern France, with research operations planned to start in 2034, and first deuterium-tritium plasma operations scheduled for 2039. In addition to ITER, several other public experiments are underway around the world, including the National Ignition Facility (NIF) in the US, the joint EU/Japan JT-60SA in Japan, the Experimental Advanced Superconducting Tokamak (EAST) in China, and the Wendelstein 7-X (W7-X) in Germany. These experiments aim to advance fusion

³ See, for example, Figure 18 in [Wurzel and Hsu \(2022\)](#); net energy gain Q as defined here is identified as Q_{sci} in the reference.

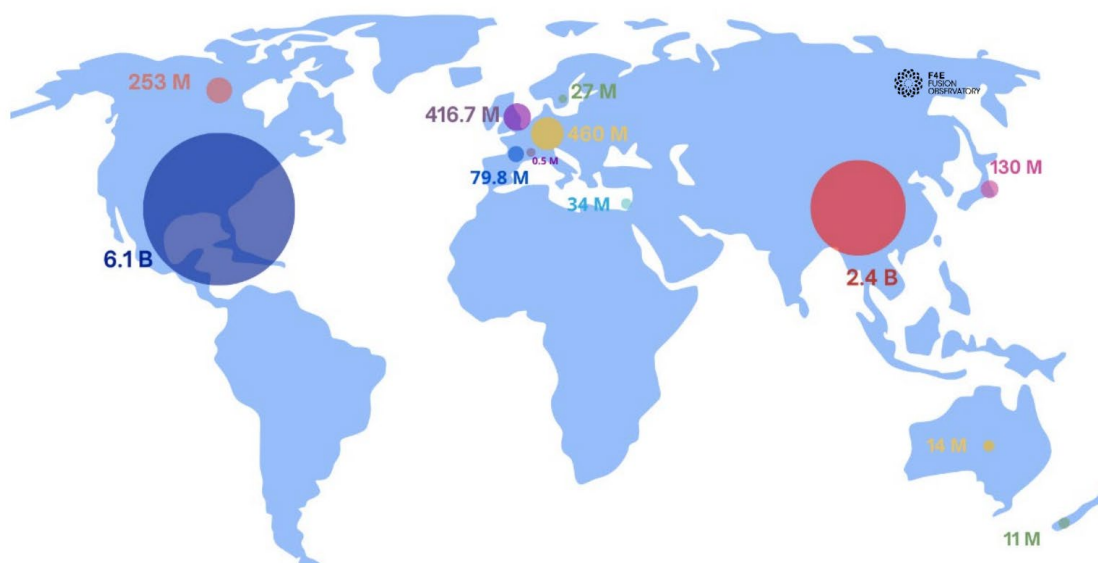
⁴ See discussion of possible timelines in foresight studies published by the [European Commission](#) and the [United States Government Accountability Office](#).

physics and develop the technologies needed for a commercial fusion power plant. Private companies working on commercial fusion solutions have also started to emerge.

Research in fusion energy is gaining momentum globally, marked by a significant increase in public and private investment. Major economies now recognise fusion as a strategic pillar of their future, post-2050, clean energy mix.

As mentioned, the EU is the leading stakeholder in the development of ITER. Europe has taken the lead in this project, with a 45 % stake in the construction costs, of which 80 % is funded from the EU budget and 20 % by France as the ITER host country (the other ITER Members share is around 9 % each).⁵ This cost distribution will change in the operation phase, with Europe providing 34 %.

Figure 1 – Geographical global investment in private fusion start-ups (€)



Source: [F4E](#).

European industry has already benefited from over €6.8 billion in contracts related to ITER and other fusion projects.⁶

In parallel to their collaboration on the ITER project, the EU and Japan have been working together under a separate agreement since 2007, referred to as the Broader Approach. It covers cooperation on three fusion-related projects, all located in Japan, aiming to complement ITER and accelerate the development of fusion power. As part of these collaboration projects, the largest and most advanced Tokamak in the world, JT60-SA, was inaugurated in Japan in 2023.

The France 2030 initiative allocated €500 million dedicated to direct subsidies for fusion and fission start-ups.⁷ Germany intends to make over €1 billion available by 2028 to go towards achieving a

⁵ See A. Delasnerie, [ITER financing by the EU budget – state-of-play](#), Policy Department for Budgetary Affairs, European Parliament, 2023, for a more detailed look.

⁶ See Fusion for Energy, [Global Investment in the Private Fusion Sector](#), 2025; this may be as high as €7.5 billion as of date of publication, according to an informal communication from the European Commission.

⁷ The [initiative](#) totals €1.2 billion in total, covering both fission and fusion.

fusion power plant by 2040.⁸ The United Kingdom aims to invest over GB£650 million to develop STEP (Spherical Tokamak for Energy Production) and upgrade research infrastructure. In the UK, Hydrogen-3 Advanced Technology (H3AT) is focusing on tritium breeding, processing, distribution, storage, recycling and disposal, which is crucial for fusion energy development. Japan's Acceleration Roadmap aims to develop a net-energy fusion demonstrator by 2035 (moved forward from 2050).⁹

The United States invested US\$1.48 billion in public funding in 2024, including US\$690 million for inertial fusion (via the part of the nuclear weapons programme dedicated to research on inertial fusion ignition) and US\$180 million for public-private R&D (Fusion Innovative Research Engine).¹⁰ In addition, it plans to invest US\$350 million in a programme of public-private partnerships to develop fusion power plants and allow for a tax credit for fusion prototypes under the Inflation Reduction Act, which might now be at risk.¹¹ The 10-year US Department of Energy plan and public-private partnerships goal had been to deploy pilot projects in the coming years, aiming at commercialisation by the end of the 2030s or early 2040s.¹²

China is planning to invest US\$1.5 billion per year in key fusion projects, such as the experimental record-holding tokamak (EAST); the publicly funded net-energy tokamak (BEST), which aims for 2027; and a large-scale demo reactor (CFETR), which targets the 2030s. In addition, state and private investment is taking place in Neo Energy, Energy Singularity and other projects.¹³

Interestingly, private investment in such risky technology development (with very low technology-readiness levels) is rising, with equity investment in fusion companies increasing substantially. The past few years have seen the rise of multiple start-ups using different fusion energy approaches and fuel choices, with over US\$8 billion in total raised to date. Some of the main, best funded, players include: Commonwealth Fusion Systems, a Massachusetts Institute of Technology spin-off, which is constructing a tokamak demonstrator with the goal of being the first such reactor to achieve net energy gain, and who recently signed a power purchasing agreement (PPA) with Google for the 2030s; Helion Energy, which has also already signed a PPA with Microsoft for 2028; and TAE Technologies, with a long track-record of pursuing a novel FRC approach. While EU start-ups are not as well capitalised, they can draw on extensive European research experience on the topic. Some examples include Marvel Fusion, which is pursuing the development of laser-based inertial confinement; Proxima Fusion, a spin-out from the Max Planck Institute capitalising on its experience with stellarators. Gauss fusion, a venture between German, French, Italian and Spanish ITER suppliers, is also working on stellarators, as well as France-based Renaissance Fusion. The Swedish Novatron also focuses on MCF, but uses a different approach – magnetic mirrors.

⁸ See for example this [account](#) of the plan. There is some [uncertainty](#) about the total amount of extra funding made available, and it is possible the quoted figure includes previously existing programmes.

⁹ See *Science Japan*, [National fusion strategy revision plan revealed](#), 16 June 2025, for an English language account.

¹⁰ See the US Department of Energy [webpage](#) on budgets for more details. The latest budget for fiscal year 2026 includes US\$745 million directly for fusion energy sciences, but important facilities like the National Ignition Facility are dual-purpose and will be supported by budget lines dedicated to the maintenance of the US nuclear arsenal.

¹¹ It is unclear whether fusion was included in the [carve-out](#) for advanced nuclear in the latest US budget bill.

¹² US Department of Energy, [Fusion Energy Strategy 2024](#).

¹³ See for example G. Conroy, [‘Inside China’s race to lead the world in nuclear fusion’](#), *Nature*, Vol. 632, 2024, p. 968.

Figure 2 presents the increase in equity investment in fusion companies by country.

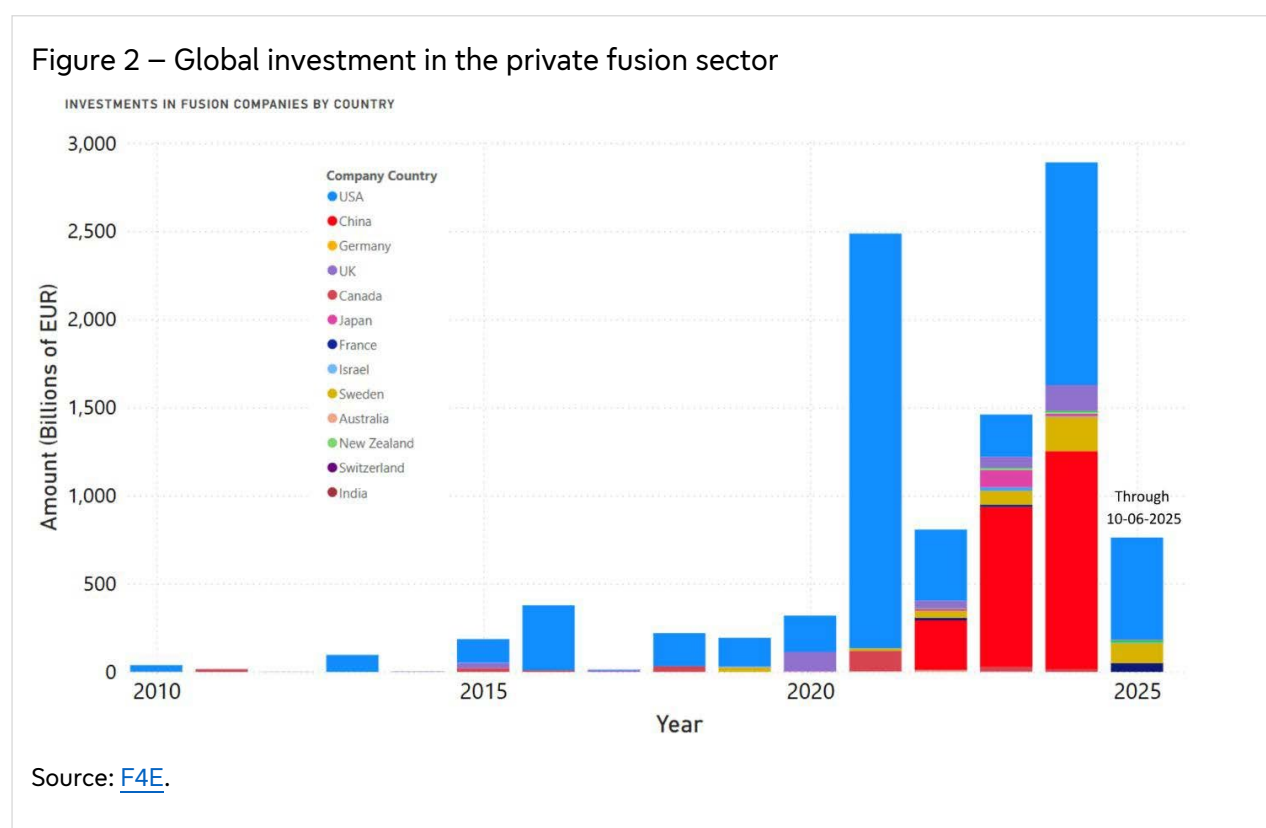


Table 1 presents a summary of the current known key players in fusion energy, including the technology used, e.g. magnetic confinement fusion (MCF), inertial confinement fusion (ICF), field-reversed configuration (FRC), and other types and approaches.

Table 1 – Organisations working on fusion energy

Organisations	Type/Approach	Location	Public/Private	Target	Funding ¹⁴
ITER	MCF/Tokamak	International	Public	Advanced research operation by 2034	€30bn+
DEMO	MCF/Tokamak	International	Public	Early design phase (300–500 MW grid connected and closed fuel-cycle)	n.a.
JET (Eurofusion)	MCF/Tokamak	EU/UK	Public	Advanced research – model for ITER (decommissioned)	n.a.
Wendelstein 7-X	MCF/Stellarator	Germany	Public	Advanced research	€1.44bn
DTT	MCF/Tokamak	Italy	Public/Private	Advanced research	€744mn
Proxima Fusion	MCF/Stellarator	Germany	Private	Commercialisation in the 2030s	€200mn
Gauss Fusion	MCF/Stellarator	Germany	Private	Pilot fusion power plant operational by the early 2040s	€27mn

¹⁴ Estimated amounts as of July 2025; updated January 2026.

Novatron	MCF/Magnetic mirrors	Sweden	Private	Commercialisation in the 2030s	€20mn
Renaissance Fusion	MCF/Stellarator	France	Private	Commercial by the early 2030s	€49mn
Marvel Fusion	ICF/Pulse	Germany	Private	Fusion support tech (laser)	€385mn
Focused Energy	ICF/Laser	Germany	Private	Pilot plant by 2035	€180mn
First Light Fusion	ICF/Amplifier	UK	Private	Fusion support tech (amplifier)	€96mn
Tokamak Energy	MCF/Tokamak	UK	Private	Net energy output in the 2030s	€308mn
Commonwealth Fusion Systems (CFS)	MCF/Tokamak	USA	Private	Net energy by 2027; 400 MW grid connected by early 2030s(ARC)	€2.9bn
General Atomics DIII-D	MCF/Tokamak	USA	Public/Private	Advanced research	€495mn
JT-60SA	MCF/Tokamak	EU/Japan	Public	Ongoing advanced research operation	€560mn
LHD	MCF/Heliotron	Japan	Public	Advanced research	n.a.
Helical Fusion	MCF/Heliotron	Japan	Private	Net power-generating prototype reactor by 2034	€21mn
KSAR	MCF/Tokamak	Korea	Public	Ongoing advanced research operation	n.a.
Type One Energy	MCF/Stellarator	USA	Private	Prototype machine by 2027	€177mn
NIF (LLNL)	ICF/Laser	USA	Public	First net gain 2022	n.a.
Helion Energy	FRC/Magneto-inertial	USA	Private	50 MW grid connected in 2028	€1bn
TAE Technologies	FRC/Beam driven	USA	Private	DaVinci power plant in the early 2030s	€1.6bn
Zap Energy	FRC/Z-Pinch	USA	Private	Commercial by the early 2030s	€331mn
General Fusion	FRC/Magneto-inertial	Canada	Private	Commercial by the early to mid-2030s	€397mn
Pacific Fusion	FRC/Magneto-inertial	USA	Private	Early development; in collaboration with General Atomics	€582mn
EAST	MCF/Tokamak	China	Public	Ongoing advanced research operation	€810mn
BEST	MCF/Tokamak	China	Public	Bridges EAST and CFETR; estimated completion in 2027	€1bn
CFETR	MCF/Tokamak	China	Public	Steady state operation of 200 MW in the 2030s	n.a.
Energy Singularity	MCF/Tokamak	China	Private	Early 2030s: Q>1 End 2030s: demo for economic fusion power	€325mn

Neo Fusion	MCF/Tokamak	China	Public/Private	Associated with the BEST tokamak	€1.8bn
China Fusion Energy (CFEC)	MCF/Tokamak	China	Public	State consolidation of fusion projects	€2bn

Data source: Adapted from L. Lundy, [Nuclear Fusion: The State of Play](#), November 2024; European Commission [foresight study](#); [Fusion Energy Base](#); the [Global fusion industry in 2025](#) report by the Fusion Industry Association; and information from organisations' websites.

3.2. Scaling fusion energy

As previously presented, the timelines for commercial fusion energy differ significantly. National and international bodies, as well as private companies, offer contrasting forecasts for prototypes, grid-connected pilot demonstration plants, and large-scale commercialisation.

Public initiatives, including the EUROfusion roadmap, largely aligned with ITER's scientific milestones for pilot plants such as DEMO expected post-2050. However, this picture seems to be changing. The proposal recently released by the European Commission for the new Horizon Europe 2028-2034 programme includes, as an example of a possible 'moonshot' project, achieving electricity production from fusion by 2034.¹⁵ The US aims to have a pilot fusion plant for the 2035-2040 timeframe, and commercial deployment from 2040. The UK's STEP programme plans a prototype by 2040. China, Korea, and Japan aim for pilot plants by 2040, with commercialisation from 2050. Companies like Helion, Commonwealth Fusion, and Type One Energy plan demonstration plants before 2030 and pilot plants in the early to mid-2030s.

The public sector is generally more cautious regarding timelines. Private companies promise faster progress to secure investor confidence, but face technical and cost uncertainties. However, if the fusion energy industry achieves early breakthroughs this could sideline some of the public initiatives, such as the DEMO project, the next step after ITER.

A critical point when trying to estimate the scaling up of fusion energy is that multiple technologies are being explored simultaneously (including tokamaks, stellarators, inertial confinement fusion, and magneto-inertial fusion, among others). While as a consequence this maximises the chances of a breakthrough, the diversity of approaches being pursued also creates uncertainty about which technology will ultimately prevail, making it challenging to predict the timeline and the future cost of electricity produced in fusion plants.

The European Commission's 'Foresight study on the worldwide developments in advancing fusion energy' presents four scenarios that illustrate the uncertainty surrounding the development of fusion energy, and help make sense of the possible pathways for the scaling up of fusion energy:

- **EUROfusion roadmap (base case):** Fusion energy development advances under public leadership, with ITER and DEMO progressing cautiously. Net energy gain is achieved in the 2040s and grid-connected pilots arrive in the 2050s, with commercial plants following in the 2060s.
- **Collaboration scenario:** Public and private sectors cooperate and early breakthroughs from start-ups help accelerate timelines, while public-private partnerships ensure safety and societal benefit. Fusion energy commercialisation starts by 2050.

¹⁵ See list of moonshot examples in European Commission, Proposal establishing Horizon Europe, the Framework Programme for Research and Innovation, for 2028-2034, [COM/2025/543](#), 2025, pp. 3-4.

- **Private 'leapfrog' scenario:** Private companies achieve early breakthroughs ahead of public programmes. They drive the first grid-connected fusion plants, potentially reshaping market dynamics and sidelining public projects. The first commercial power plants are rolled out in the 2040s.
- **Fusion is hard (pessimistic):** Obstacles to fusion are greater than originally envisioned, and neither public nor private initiatives are able to overcome the key challenges in the planned timeframes. Timelines are seriously delayed. Commercial fusion energy is pushed back to the 2070s.

Given the overall state of the technology, current roadmaps from the different players and their own risk assessment, this foresight study sees the second scenario (collaboration) as the most likely. Relative to the EUROfusion baseline, ITER would maintain an important role, but DEMO would be sidelined as private companies would be able to achieve comparable milestones within a faster timeframe.

4. Key challenges

Despite the dynamism and optimistic outlook currently prevalent in much of the fusion sector, significant challenges remain on the road to commercial fusion energy. Importantly, many of these are technical in nature, with several key enabling technologies still at low levels of technological development. While public projects looking into solutions to these are ongoing, their timelines do not always fit with many of the private players' roadmaps. Beyond the technical validation of fusion energy, additional challenges will also need to be overcome to deploy fusion as a commercially viable electricity source. The key challenges are summarised below.

4.1. Technical performance

Several important technical obstacles remain, given there have been few opportunities to validate proposed solutions in practice. These range from the behaviour of burning plasma (where most of the heating comes from the fusion process), to key enabling technologies linked to materials and the fuel cycle:

- **Plasma control:** one of the main challenges in the history of fusion has been plasma control, given that plasma is prone to instabilities and turbulence. This is a particular problem for MCF, where maintaining the plasma in a steady state under confinement for lengthy periods is a critical step towards efficient power production. Furthermore, the burning plasma regime has yet to be properly studied experimentally. This is an area where the use of AI promises to have an impact, as might advanced reactor types like the stellarator.¹⁶
- **Blanket:** the inside blanket of a MCF reactor is in a harsh environment, with the inner walls exposed to significant thermal and neutron radiation. On top of this, the blanket will also be responsible for neutron shielding, tritium breeding and heat extraction, while requiring accessible maintenance and ease of replacement. Blankets are a key component of future reactors, and research into the best and most resilient materials is an ongoing task. In the EU, the IFMIF-DONES facility is being built specifically to conduct research into such materials but is only scheduled to be fully online in 2035. Some private initiatives are proposing novel approaches that would avoid some of

¹⁶ For more details, the original scientific article is J. Degraeve et al., '[Magnetic control of tokamak plasmas through deep reinforcement learning](#)', *Nature*, Vol. 602, 2022, p. 419. Google Deepmind's [blogpost](#) on the subject gives a more accessible account.

these issues, such as the use of liquid metal, but these pose their own challenges and still require experimental validation.

- Tritium breeding: with deuterium-tritium the most widely planned fuel mix, sourcing the necessary tritium is a critical issue, given that it is exceedingly rare in nature and only produced for civilian purposes in very limited quantities at a handful of nuclear reactors worldwide. A new facility for tritium separation is under construction in Romania but will only be sufficient for research and development purposes, not full commercial operation.¹⁷ The proposed solution to this issue is to use the neutrons generated by the fusion reaction to create the necessary tritium inside the reactor, via fission of lithium in a breeding blanket. However, this solution has not yet been extensively tested in practice, and there are questions whether it will be possible to produce sufficient tritium not only to feed the reactor, but also to have reserves available to restart it.¹⁸ While ITER is planned to test solutions for tritium breeding to advance the technology, private companies are calling for research to be accelerated.¹⁹ Overall tritium reserves will also have an impact on the speed of expansion of commercial plants, given the need for a start-up stock for each new one.

4.2. Hardware challenges

Magnets (in MCF) and lasers and targets (in ICF) are the key hardware components of future fusion reactors. There have been considerable developments in magnet technology recently, with private companies such as CFS pushing research into the use of high-temperature superconducting magnets for fusion – for a given size, these allow for significantly stronger magnetic fields than the traditional superconducting magnets being used at ITER. This is part of the reason why the reactor currently under construction by CFS is aiming to achieve net energy gain, while being considerably smaller than ITER. However, there have also been claims that there may be significant structural issues with more compact reactors using high-temperature superconducting magnets.²⁰

With net energy gain demonstrated at the NIF, the scaling up of ICF approaches would involve two main areas. Despite the net energy gain experiments, laser energy efficiency is very low at the NIF, with the lasers only depositing a few percent of the original electrical energy used to fire them onto the target. While laser technology is progressing rapidly thanks to broader applications, the required improvement of about an order of magnitude still poses a considerable challenge. The other aspect requiring significant progress is linked to shot frequency and target construction: as NIF is an experimental facility, it can do two or three shots per day, but a commercial power plant would involve at least a cadence of around a shot per minute. This would not only require considerable development into target handling and laser cooling, but also in construction, to considerably lower target costs.

4.3. Lifecycle issues

Even if less relevant at the current stage of development, lifecycle issues will assume greater importance as work progresses towards pilot plants, and particularly so for any commercial design once that stage is reached. Key questions are related to: reliability and maintenance, especially in

¹⁷ See Nuclearelectrica, [press release](#), 2 June 2025. This facility will work in tandem with Romania's CANDU fission reactors. The limited number of reactors of this type worldwide limits the possibilities for expanding this process.

¹⁸ D. Clery, 'Out of gas', *Science*, Vol. 376, American Association for the Advancement of Science, 2022, p. 1372.

¹⁹ Proxima Fusion's [position](#) on the subject is an example.

²⁰ G. Federici et al., 'Relationship between magnetic field and tokamak size—a system engineering perspective and implications to fusion development', *Nuclear Fusion*, Vol. 64, IOP Publishing, 2024, 036025.

situations such as in MCF, where the inside of the reactor will be a hot, radioactive environment due to leftover heat and neutron activation, requiring remote handling; durability, linked to the materials issues discussed above but particularly important for commercial operations; and eventually decommissioning, particularly for those cases where parts of the reactor have become radioactive due to neutron activation.

4.4. Construction challenges and skills

Although already relevant in some cases, especially in terms of possible skills shortages, challenges related to construction and skills availability will come into their own for the deployment of commercial plants. These are related to the build-up of the supply chain, including possible concerns about critical materials, and the availability and retention of skilled staff. Although not immediately problematic, some of these aspects (such as having a pipeline of adequately trained, PhD-level plasma physicists) require long lead times, so proper planning must start well in advance. In addition, as shown by the experience of ITER, plants are incredibly complex pieces of advanced engineering, the construction of which presents its own set of challenges. Ensuring adequate build-up of experience and sharing of lessons learned will be an important part of any scale-up.

4.5. Regulatory environment

No EU legal framework specifically addresses fusion energy. The EU and national nuclear safety rules, under Euratom, were designed for fission reactors and do not fully account for fusion energy's unique characteristics.²¹ Fusion reactors have no self-sustaining chain reaction or meltdown risk and create no high-level long-lived waste.²² Some national authorities opt to adapt fission-based or radiation protection laws to fusion projects. For example, the ITER facility in France was licensed under the existing nuclear legal framework for basic nuclear installations (*Installation Nucléaire de Base* – INB). Others, such as the German Wendelstein 7-X fusion experiment, are licensed under radiation protection law (*Strahlenschutzgesetz*), similar to licensing for particle accelerators. The European Commission study on the applicability of the regulatory framework for nuclear facilities to fusion facilities, recommended coordinated EU action on establishing a fusion-specific legal and regulatory framework, building regulatory capacity, and harmonising safety requirements across Member States.

4.6. Funding and economic viability

As a complex technology at the cutting edge of current possibilities, fusion requires considerable investment. For example, ITER will cost in excess of €30 billion, while private demonstrators are often estimated to come in at around €1 billion, with later commercial reactors possibly costing multiples of that figure. Although the success of future fusion developments may affect overall funding levels, and how they impact perceptions of the viability of the technology, public and private initiatives face different challenges where funding is concerned. While funding for ITER and other current experiments is relatively secure, funding for projects such as DEMO will be much more politically exposed to future developments. This creates the risk of a situation in which public funding is cut because the private sector, having reached some important milestones first, is perceived to be ahead, but nevertheless remains dependent on continued research on critical

²¹ See for example European Commission, [Study on the applicability of the regulatory framework for nuclear facilities to fusion facilities](#), 2021.

²² For more details, see for example the information [page](#) of ITER or the International Atomic Energy Agency [FAQ](#).

systems from the public side. The key challenge for the private sector will be to raise the significant sums that will be needed to build the first demonstrators, and especially first prototypes. While success will breed success, the converse is also true, and there is a risk that a failure by one company will have an impact on others pursuing similar approaches. Finally, as is usually the case with new technologies, the first commercial reactors will likely be very expensive, and need public support to build.

The future economic viability of fusion is hard to predict, being dependent not only on the ultimate cost of fusion electricity when it is put on the grid, but also on the evolution of the electricity market, including demand, and other technologies such as small modular reactors or geothermal plants. Some of the private start-ups have advanced estimates of a future levelised cost of electricity (LCOE) that would be competitive with other sources of firm (non-intermittent) power, such as nuclear or gas with carbon capture and storage. However, too much uncertainty remains regarding cost factors, even technology related issues such as maintenance, downtime, and the total lifetime of a fusion plant. However, the main advantages of fusion, as a spatially compact, low carbon, stable power source with abundant fuel that does not suffer the problems of nuclear power, are an argument that there should be room for it in future power mixes.

5. Policy considerations

Fusion energy has been a strategic priority for the EU's long-term decarbonisation and its energy security. The Euratom Treaty and EU research and innovation framework programmes have contributed to building a research ecosystem around fusion energy.

The Draghi report states that nuclear fusion is a disruptive technology that holds the potential to revolutionise the energy landscape in the second half of this century.²³ The report also mentions that the ITER project has propelled the EU to the forefront of global fusion research and calls for a stable and predictable fusion ecosystem for industrial innovation to be created that leverages the ITER project while ensuring a clear technology development roadmap.

5.1. EU fusion policy

The EU multiannual financial framework (2021–2027) funds ITER, with a budget of €5.61 billion. The Fusion for Energy Joint Undertaking (F4E) manages the EU's participation in ITER. The Euratom Research and Training Programme (2021–2025), contributes €583 million for indirect action in fusion research and development. In addition, Horizon Europe (2021–2027) funds many complementary enabling technologies (e.g. AI, materials science) and supports fusion-relevant innovation. For example, the European Innovation Council accelerator programme awarded €2.5 million, and an opportunity for a €15 million investment in Proxima Fusion, and the European Institute of Energy and Technology's Innoenergy partner supported Novatron Fusion in securing €5 million in seed funding.²⁴

The European Commission manages these funding programmes, including F4E, which deals with the engineering and industrial delivery for ITER. EUROfusion complements this work by implementing the EU's fusion research roadmap. EUROfusion is a consortium of national laboratories and research institutes across Europe (including the UK and Switzerland). From 2021 to 2025, EUROfusion

²³ See M. Draghi, [The future of European competitiveness](#) – Part B: In-depth analysis and recommendations, European Commission, 2024, p. 21.

²⁴ See [Proxima Fusion](#) and [Novatron Fusion](#), press releases.

receives €549.4 million in support from the Euratom research and training programme (2021–2025), with about €450 million in funding matched by the Member States.

EUROfusion's 2018 'European research roadmap to the realisation of fusion energy' outlines the conventional view of how Europe intends to transition from experimental fusion science to grid-connected electricity generation from fusion by the 2050s. It provides a step-by-step approach and prioritises investments to achieve this goal. The three-core pillars of the roadmap are:

1. ITER: demonstrate the scientific and technical feasibility of fusion at reactor scale.
2. IFMIF-DONES: qualify materials for future fusion reactors by simulating fusion-relevant neutron environments.
3. DEMO: build the first fusion power plant prototype that produces net electricity for the grid.

In addition, research and development of stellarators and tritium technologies is to be carried out in parallel. Today, EU policy is currently in a state of flux, with fusion by 2034 presented as a possible 'moonshot' project in the 2028–2034 Horizon Europe programme, and a specific strategy on fusion expected in late 2025.

The current roadmap is based on the success of ITER, and its implementation is therefore closely monitored. The 'Interim evaluation study of the implementation of the Council Decision (Euratom) 2021/281' assessed Euratom's contribution to the ITER project through F4E between 2021 and 2023.²⁵ An important conclusion of this study is the need for a comprehensive EU fusion strategy to improve coordination of the fragmented landscape of fusion initiatives. It also states that, while the ITER project remains scientifically relevant and yields significant socio-economic benefits for the EU, these gains might be lost due to escalating costs and persistent delays. The evaluation study proposes greater flexibility, deeper integration between ITER and F4E, and robust knowledge management to preserve the industrial expertise gained. It also suggests that the EU's continued involvement is key. ITER's scale requires resources and expertise beyond what individual Member States can provide and, therefore, demonstrates the added value of a coordinated EU approach.

Another recent European Commission study, 'Analysis on a strategic public-private partnership approach to foster innovation in fusion energy' also supports a clear and overarching EU fusion strategy and vision. In addition, it also highlights a significant risk that the EU's extensive fusion know-how and industrial value chain, developed through decades of public investment in projects such as ITER, could be lost as ITER nears completion. The study cautions that the risk is compounded by increasing international competition and the growing global private fusion sector, which could draw investment and expertise away from the EU. The study suggests that promoting faster, viable commercialisation of fusion energy requires both public and private investment. To address these challenges and maintain EU leadership and competitiveness in fusion, the study strongly recommends: i) establishing a strategic public-private partnership (PPP) framework – a co-programmed European partnership; and/or ii) a F4E innovation partnership; and iii) making fusion energy eligible for funding under complementary sources/instruments (e.g. Innovation Fund, InvestEU, Connect Europe Facility, European Investment Bank). The study underscores heavily the importance that the EU engages constructively with research and industry stakeholders in a joint effort to leverage the EU's scientific leadership in advancing fusion energy.

The High-level European roundtable on fusion that met in March 2024, asked the Commission to foster closer collaboration between public and private entities, attracting investments in Europe and

²⁵ European Commission, C. Pancotti, G. Catalano, L. Colnot, S. Banfi, [Interim evaluation study of the implementation of the Council decision establishing the European joint undertaking for ITER](#), 2025.

developing regulatory frameworks.²⁶ More recently, the European Commission's Fusion Expert Group released an opinion to support the development of an EU fusion strategy.²⁷ It calls for continued support for ITER while maximising its benefits to the EU fusion ecosystem, and for the development of a comprehensive roadmap towards a pilot plant, while giving due consideration to the full suite of required technologies and supporting their development, when necessary, possibly through international cooperation. It also recommends supporting alternative pathways such as ICF, investment in infrastructure, strengthening the supply chain, and developing a workforce strategy. Many of the recommendations deal with two key aspects: establishing an EU-wide regulatory framework for fusion, and integrating, coordinating and creating synergies with private initiatives. The latter would be achieved through two main vehicles, a public-private partnership and a European fusion stakeholder platform, which would join existing actors in a revised and streamlined governance structure. The paper is comprehensive in its approach but sticks closely to the vision reflected in the EUROfusion roadmap – an ambitious, flexible outlook, with stronger private sector participation, may be needed if Europe is to remain competitive and able to respond to developments abroad.²⁸

Table 2 – Key stakeholders in the EU fusion energy sector

EU fusion energy stakeholders	Roles
European Commission	Defines strategy, oversees Euratom, negotiates international agreements
ITER	International Thermonuclear Experimental Reactor
Fusion for Energy (F4E)	EU agency managing ITER procurement and industrial engagement
EUROfusion	Consortium of 30+ national labs; coordinates fusion research and roadmap delivery
European Fusion Association	Represents EU fusion value chain-technology providers, includes US start-ups
Fusion Industry Association	International fusion industry association, includes several EU fusion start-ups
Council and European Parliament	Approves work programmes and funding envelopes for fusion energy R&I

Beyond research and industry stakeholders, it is also important to consider the public perception of fusion energy. Recent surveys found that Europeans are largely favourable, with a majority of the public expressing support. However, this support seems to be influenced by existing attitudes towards nuclear energy, science, and technology and perceptions regarding feasibility, development costs and timelines. There are also hints of an emerging fusion 'brand', distinct from that of nuclear, which is associated in the public consciousness with concepts of endless, clean energy and scientific development.²⁹

²⁶ For the proceedings of the roundtable, see European Commission, L. Capisani (editor), C. Camus (editor) and M. Papadopoulou (editor), [Fostering fusion innovation – High-Level European roundtable](#), 2024.

²⁷ European Commission, [Fusion Expert Group opinion paper – Towards the EU fusion strategy](#), 2025.

²⁸ See for example stakeholder feedback in [F4E's stakeholder survey](#).

²⁹ For two recent studies on public support for fusion in Europe, see C.R. Jones et al., [‘The clock is ticking: Understanding the ‘mixed feelings’ about fusion energy in Europe’](#), *Energy Research & Social Science*, Vol. 113, Elsevier, 2024, 103538, and C. Oltra et al., [‘Public acceptance of fusion energy in Europe’](#), *Fusion Engineering and Design*, Vol. 222, Elsevier, 2026, 115430. For the concept of a fusion brand, see for example the discussion in C. Turcanu et al., [‘Fusion energy: A deeper look into attitudes among the general public’](#), *Fusion Engineering and Design*, Vol. 161, Elsevier, 2020, 111891.

5.2. EU fusion policy issues

From the studies mentioned above and the stakeholders' dialogue conducted in the 'Foresight study on the worldwide developments in advancing fusion energy', several key EU fusion policy issues can be identified. These issues also emerged in recent workshops on fusion energy organised by Members of the European Parliament.

Strategic direction and technology diversity

- **The EU lacks a comprehensive fusion energy strategy.** The lack of a comprehensive strategy coordinating public research efforts with emerging private-sector fusion initiatives may delay knowledge transfer, infrastructure sharing, and joint innovation across the ecosystem. The European Commission is preparing a fusion strategy, expected in Q4 2025, which should address this gap through clearer public-private co-governance and resource alignment. A public call for evidence on this strategy took place in June 2025.³⁰
- **Fusion R&D is overly concentrated on a single technological pathway.** There is increasing optimism and a growing anticipation among researchers and engineers that one or more fusion approaches will succeed by 2040–2050, though it is unclear which one (e.g. tokamaks, stellarators, or inertial fusion).³¹ So far, EU public funding has mainly focused on supporting tokamak-based fusion, while private initiatives in the EU are working on a range of other technologies. The focus on a single approach could limit impact, if alternative concepts prove more viable or faster to deploy. Additionally, the lack of overlap between public and private initiatives may limit possibilities for developing synergies.
- **Fusion is absent from key EU energy and climate long-term planning tools.** Several fusion energy pilot projects could become operational before 2050. Similar to other emerging technologies that could have a significant post-2050 impact (e.g. small-modular reactors, direct air capture and carbon capture and storage), fusion can potentially contribute to reducing greenhouse gas emissions, the expected strong electrification of the global economy and to addressing future increased demand for electricity from new technologies such as AI.

Governance, regulation, and public-private partnerships

- **Limited coordination between public and private stakeholders.** EU public and private fusion stakeholders do not appear to be engaging efficiently. While the public programmes (e.g. ITER and EUROfusion) take a long-term approach, private companies appear to be fast paced, but both aim at developing cost-effective and reliable fusion power. Fragmentation also affects the private sector, with the supply chain and start-ups having different needs and competing industry associations. The EU's leadership in fusion would benefit from a stronger, structured partnership between public institutions and all private actors, as well as a shared strategic vision.
- **Fusion energy is usually regulated under frameworks designed for nuclear fission,** despite the significant differences in safety and operational risks.³² Although most EU

³⁰ See the Commission's [page](#) of the call for details, including links to the submissions.

³¹ For examples reflective of such opinions, see F4E's [Stakeholder survey analysis on the FEG opinion paper](#), 2025, especially the section on 'Market entry and commercialisation', p. 22.

³² European Commission, L.-G. Eriksson (editor), R. Passalacqua (editor) and C. Ibbott (editor), [Exploring regulatory options for fusion power plants](#), 2021.

nuclear frameworks include and apply a risk-proportionate approach, some believe this has caused delays in ITER.

- **Intellectual property (IP) governance is not yet clearly defined.** Public programmes have traditionally operated under open-access or shared-knowledge principles. As private investment increases, stakeholders seek stronger IP protection, creating tension and the need for a balanced public-private IP framework, as well as the development of international standards.

Financing and industrial development

- **Funding for EU startups and projects remains uncertain.** Industry stakeholders feel that while funding for ITER and EUROfusion is relatively secure, financial commitments towards longer term projects are unclear. This applies to public projects like DEMO but is also the case with later start-up funding rounds. In addition, the lack of acknowledgement of the potential of fusion energy in green finance or industrial strategies increases investors' perceived risk. Currently, EU start-ups have lower levels of (private) funding compared with their US and Chinese counterparts, although they have been competitive in early funding rounds.³³
- **Fusion supply chains can potentially face critical dependencies.** As ITER is demonstrating, fusion energy systems rely on complex supply chains. Many of the required materials and components are subject to global competition, geopolitical risks, or dual-use constraints. EU fusion supply chain and industrial capacity is building up, but continuous support is needed.
- **Europe risks lagging behind in the global fusion race.** The EU traditionally leads the large-scale projects like ITER and continues to see considerable public investment in fusion. However, it has comparatively fewer private start-ups, which have raised less funding so far. With the race towards commercialisation eventually shifting to the private sector, the EU risks lagging behind despite its track record.

Skills, talent, and research capacity

- **The fusion talent pipeline is limited.** Fusion energy research and industrial capacity to deliver over the next two decades is dependent on the supply of skilled scientists and engineers, with talent retention playing a potentially important role. Skills shortages may become acute at the commercial deployment stage, unless addressed beforehand.
- **Advanced digital capabilities are key.** Fusion research and development increasingly uses AI, high-performance computing (HPC) and digital twins. These tools are essential for the design, real-time operation and maintenance of fusion energy prototypes.

Public acceptance and ethics

- **Energy fusion is often confused with nuclear fission.** Although there seems to be public support, fusion energy remains poorly understood by citizens and is often confused with nuclear fission, which potentially creates acceptance barriers for the upcoming fusion energy prototypes and first-of-a-kind pilot projects.
- **Dual-use and spillover effects raise ethical considerations.** Research in fusion energy is contributing to the development of advanced materials, AI, cryogenics, and

³³ For some statistics, see F4E, [Global Investment in the Private Fusion Sector](#), 2025. Regarding initial funding rounds, Proxima Fusion for example raised €130 million in series A funding, which is similar to the initial series A funding of US\$115 million for CFS, the best-funded US start-up. However, that round dates back to 2019 and CFS has raised close to US\$3 billion in [total](#) as of publication.

medical technologies, including other dual use technologies (e.g. lasers, magnets and railguns). This might raise ethical concerns, particularly when fusion innovation is applied beyond its energy applications.

Sustainability, cost, and market integration

- **Limited information about environmental sustainability.** Although fusion energy is often promoted as a clean and safe alternative to fission, lifecycle analysis of its environmental impacts is limited. Key concerns include radioactive waste from neutron activation, cooling water demands, and end-of-life decommissioning.
- **Techno-economic data remains weak.** Few fusion energy developers can currently provide robust cost projections or LCOE estimates. Clearer techno-economic data is needed to integrate fusion into the EU's long-term (post-2050) energy and climate scenarios or justify policy and investment support compared to other emerging energy technologies.
- **Grid and market compatibility is uncertain.** The integration of fusion energy into future EU electricity systems poses both technical and regulatory challenges. Fusion power plants, when deployed, are expected to be large, capital expenditure intensive, and suited to provide constant baseload electricity. However, this model may not be the best fit to an EU electricity market and infrastructure which are undergoing a transformation towards greater decentralisation, flexibility, and intermittent-renewable electricity integration.

6. Way forward

Public initiatives have long dominated fusion research and development, but the current state of the technology hints that leadership in the quest for commercialisation of fusion energy will shift to the private sector in the not-too-distant future. With the different public and private timelines often in tension, the magnitude of the remaining challenges on the road to viable fusion energy production means the timing of this transition remains very uncertain. How to manage it will be a central plank of future strategic approaches to fusion. A flexible approach, striking the correct balance between public and private support, as well as the different technologies in play, will be key.

Smart public-private collaboration, properly leveraging the strengths of both, will be important to successfully develop fusion technology. On the public side, continuing support for critical projects such as ITER would allow them to fulfil their scientific value, establishing supply-chains, and reinforcing the complementarity and experience they can provide to private actors. Adequately supporting EU-based private initiatives could close gaps with non-EU competitors, although fundraising potentially remains a traditional comparative weak point that needs to be addressed. A fine-tuned regulatory environment and skilled workforce, coupled with enhanced public awareness of the benefits of fusion to facilitate acceptance, could help establish the groundwork and drive the incentives for a dynamic EU fusion sector.

Table 3 – Possible action that could help translate these goals into policy

Policy issues	Possible action
Strategic direction and technology diversity	• Monitor the development and implementation of the upcoming EU fusion energy strategy.

	• Diversify funding towards multiple fusion approaches (e.g. tokamaks, stellarators, inertial fusion), to hedge potential technological risks, via either public or private initiatives. • Integrate fusion energy considerations into EU energy and climate policy frameworks, including the Net-Zero Industry Act and the National Energy and Climate Plans (NECPs).
Governance, regulation, and public-private partnerships	• Consider establishing a public-private partnership to coordinate public and private efforts, ensuring shared governance and strategic alignment. Public-private collaboration should equally cover key enabling technologies. • Push for the development of a fusion-specific EU regulatory framework that reflects the unique safety and operational characteristics of fusion energy. • Support clear guidelines for intellectual property (IP) management in fusion, sharing data from public funded projects, while protecting proprietary technologies.
Financing and industrial development	• Revise the scope of the EU Taxonomy for sustainable activities to facilitate green financing of fusion energy prototypes and pilots. • Support the development of a resilient fusion supply chain, focusing on critical materials, such as tritium and rare earths, through the promotion of strategic investment and partnerships. • Foster international collaboration to standardise fusion technologies and maintain EU competitiveness in the global fusion race. Work with the International Atomic Energy Agency to standardise safety requirements.
Skills, talent, and research capacity	• Promote fusion energy education and training programs within existing EU programmes such as the Marie Skłodowska-Curie actions and the European Research Council. • Facilitate the integration of advanced digital tools (AI, HPC) into fusion research and operations, promoting cross-sector collaboration.
Public acceptance and ethics	• Increase citizens' understanding of fusion energy and consider improving the distinction between fusion energy and nuclear fission. • Monitor the development of dual-use concerns and ensure responsible innovation.
Sustainability, cost, and market integration	• Promote the study of lifecycle assessments of fusion energy systems to evaluate environmental impacts. • Use existing initiatives, such as the strategic energy technology (SET) plan, to monitor the fusion energy industry, including techno-economic data, business models and market integration, to inform policy and investment decisions.

7. Conclusion

Fusion energy could be vital for Europe's future. It offers the promise of a safe, clean, and virtually limitless source of power that could help it achieve deep decarbonisation beyond 2050, strengthen energy security, and support the electrification of the economy. Fusion could play an essential role alongside renewables and other technologies in the future energy mix – without the fuel supply risks or long-lived radioactive waste that come with nuclear fission.

However, the path toward fusion is uncertain. Scientific, engineering, regulatory, and financial challenges need to be overcome. But the pace is picking up: private sector interest is growing rapidly in what has long been a publicly driven field. This is manifest in two emergent visions of the path to fusion commercialisation. One looks to ITER as the key stepping stone and sees fusion as a longer-term project. The other believes rapid progress towards commercialisation is possible and relies on private investment in start-ups and their ambitious timelines. Europe is currently leading the public effort by hosting and driving the world's flagship fusion project, ITER, which contributes to the creation of a talent pool and an industrial supply chain. However, it is at risk of falling behind when it comes to private initiatives. This tension may start to become clearer in as little as two to three years, when CFS's SPARC tokamak is predicted to begin operations in the US.

To stay ahead in the development of this strategic technology, the EU must do more. There is a need for a clear vision that brings together public and private efforts with better governance, regulation and partnerships. The EU's industrial value chains need nurturing by ensuring a stable pipeline of projects and financing from both public and private sources. The talent and research capacity required to keep Europe in the lead should be mobilised while engaging with citizens, managing expectations, and monitoring progress on cost, sustainability, and market readiness.

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Fusion, regarded by many as the ultimate form of clean energy, may finally be within reach. This analysis looks at the context of the latest advances, in particular the explosion of private-sector interest. Overall, this has led to an acceleration of many planned timelines, although a lot of uncertainty remains, given the magnitude of the remaining technical challenges. In addition, questions also persist around skills, the regulatory environment, financing and economic viability. With a new EU strategy expected in late 2025, this analysis concludes with an overview of current policy, and discusses some of the challenges the EU faces in setting a pathway towards the commercialisation of fusion energy.

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